

Java Spring Boot Backend Team Nomenclature Guide

1) Project & Module Structure

- Pattern: com.<company>.<project>[.<module>]
- All lowercase for packages.
- <company>: short, lowercase org name.
- <project>: product/service identifier.
- <module>: optional feature slice.

2) Standard Packages (by layer)

- controller
- service
- service.impl (only if you split interface/impl)
- repository
- entity (JPA models)
- dto (request/response models)
- mapper
- exception
- config
- constants
- util (pure stateless helpers)
- security (filters, JWT, etc.)
- integration (clients to external systems)
- events (publishers, listeners, payloads)
- validation (custom validators)

3) Class Naming

- Controller: XxxController (REST endpoints)
- Service (interface): XxxService
- Service (impl): XxxServiceImpl
- Repository: XxxRepository
- Entity: singular PascalCase
- DTOs: XxxRequest, XxxResponse, XxxListItem, XxxPageResponse
- Mapper: XxxMapper
- Enum: XxxStatus
- Exception: XxxNotFoundException, InvalidXxxStateException
- Config: XxxConfig
- Constants: XxxConstants

4) Method Naming

- CRUD: createXxx, getXxxById, updateXxx, deleteXxx
- Queries: findBy<Field>, existsBy<Field>, countBy<Field>
- Booleans: is/has/should
- Utilities: verbNoun

5) Variables & Fields

- camelCase for variables/fields.
- SCREAMING_SNAKE_CASE for constants.
- Collections: suffix with type hint when helpful.

6) REST API Design

- Base path: /api/v<version>/<resource>
- Plural nouns for resources.
- Sub-resources: /orders/{id}/items
- Actions: /users/{id}/activate
- Query params for filters/sorting/paging.

7) Request/Response Models

- Request DTOs: minimal, validation-annotated.
- Response DTOs: safe fields only, timestamps as ISO-8601 UTC.
- Pagination: include content, page, size, totalElements, totalPages, sort.

8) Error Model & Codes

- HTTP status + stable error body.
- Error code naming: DOMAIN_REASON.

9) Database Naming

- Tables: plural snake_case.
- Columns: snake_case.
- Primary key: <entity>_id.
- Foreign key: reference_id.
- Constraints/Indexes: pk_, fk_, uk_, ix_ prefixes.

10) Git Branches & Commits

- Branches: <type>/<short-kebab-summary>.
- Commit messages: Conventional Commits format.

11) Logging

- Use Slf4j placeholders.
- No secrets/PII in logs.
- Use traceId/spanId (MDC) for tracing.

12) Testing

- Test class: XxxServiceTest, XxxControllerTest.
- Test method: methodName_expectedBehavior_condition.
- Integration test suffix: IT.
- Use test data builders.

13) Config & Profiles

- Profiles: local, dev, staging, prod.
- Files: application.yml + profile-specific.
- Properties: kebab-case keys.

14) Security Artifacts

- JWT filter: JwtAuthenticationFilter.
- Security config: SecurityConfig.
- Permission enums: Permission, roles as ROLE_*.

15) Validation & Exceptions

- Constraint annotations on Request DTOs.
- Global handler: GlobalExceptionHandler.
- Use @ControllerAdvice.

16) Events & Messaging

- Event payload: XxxEvent (immutable).
- Publisher: XxxEventPublisher.
- Listener: XxxEventListener.
- Topics/queues: kebab-case.

17) External Integrations

- Client interfaces: XxxClient.
- Config: XxxClientConfig.
- Fallbacks: XxxClientFallback.
- DTOs under integration.<provider>.dto.